

Analysis of Factors Affecting Individual Medical Expenses and Comparative Study of Prediction Models

Abstract

This study aims to explore the key influencing factors of individual medical expenses and construct an accurate prediction model. Using the data of individual medical expenses in the United States as a sample, descriptive analysis, correlation analysis, linear regression, decision tree, and random forest models were employed for the research. The results showed that smoking status, age, and BMI were the core factors affecting medical expenses. The medical expenditures of smokers were significantly higher, and high BMI and age growth would further increase the costs. Gender, region, and the number of children had relatively weak influences. Model comparison indicated that the random forest model performed the best and could better capture the nonlinear relationships between variables. The research can provide data support and methodological references for health intervention for high-risk populations, medical insurance pricing, and the optimization of medical resource allocation.

Introduction

In the context of the continuous increase in global healthcare costs, accurately identifying the influencing factors of individual medical expenses and constructing

efficient prediction models have become crucial requirements for health management, insurance actuarial science, and public health policy formulation. Existing studies mostly rely on traditional linear models, which have limited ability to capture the complex nonlinear relationships between variables and are unable to comprehensively reveal the interaction effects of factors such as smoking, BMI, and age on medical expenditures.

The data file used in this project is insurance.csv. The dataset contains 1,338 observations of individual medical expenses in the United States. The response variable is charges, and the predictor variables include age, sex, BMI, number of children, smoking status, and region.

This study takes the individual medical expense dataset as the research object. Firstly, through descriptive statistics and correlation analysis, it explores the distribution characteristics and interrelationships of each variable; subsequently, three prediction models including linear regression, decision tree, and random forest are constructed, and their performance on the test set is compared and evaluated.

Exploratory Analysis

Table1. Descriptive statistics of smoking status and medical expenses

smoker	count	mean_charges	sd_charges	median_charges	min_charges	max_charges
no	1064	8434.268	5993.782	7345.405	1121.874	36910.61
yes	274	32050.23	11541.55	34456.35	12829.46	63770.43

The grouped statistical results of smoking status and medical expenses are presented, divided into two groups based on whether one smokes. It shows the sample

size, the mean, standard deviation, median, minimum value and maximum value of medical expenses. The data clearly demonstrates that the mean and median of medical expenses for smokers are significantly higher than those of non-smokers.

Table2.Descriptive statistics of medical expenses and health characteristics for different gender groups

sex	count	mean_charges	sd_charges	mean_bmi	mean_age
female	662	12569.58	11128.7	30.37775	39.50302
male	676	13956.75	12971.03	30.94313	38.91716

Grouped by gender, the data presented the sample size, the mean and standard deviation of medical expenses, the mean BMI, and the mean age. The data shows that the average medical expenses and average BMI of men are slightly higher than those of women, while the average age of women is slightly higher. This comparison reveals the distribution differences in medical expenditures and key health indicators between different genders.

Table3. Statistics on medical expenses, BMI and smoking rates in different regions

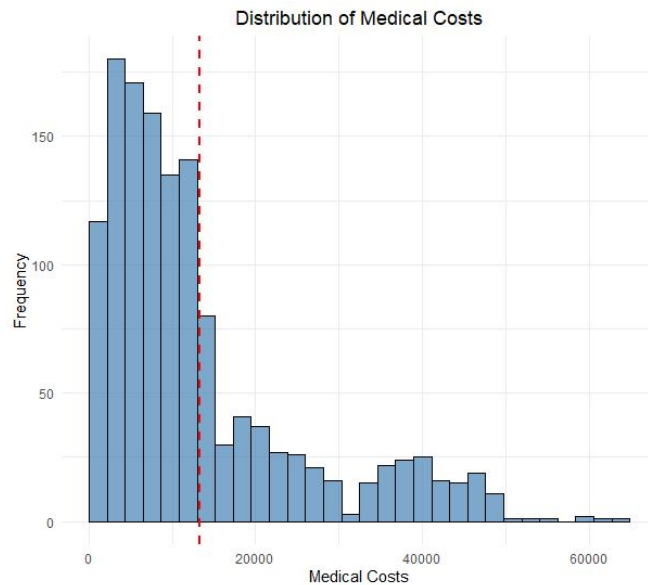
region	count	mean_charges	median_charges	mean_bmi	smoker_rate
northeast	324	13406.38	10057.65	29.1735	20.67901
northwest	325	12417.58	8965.796	29.19978	17.84615
southeast	364	14735.41	9294.132	33.35599	25
southwest	325	12346.94	8798.593	30.59662	17.84615

Grouped by the four major regions of the United States, the sample size, mean and median of medical expenses, average BMI, and smoking rate were compared. The data showed that the average medical expenses, BMI, and smoking rate in the southeastern region were the highest, and there were significant differences among the regions.

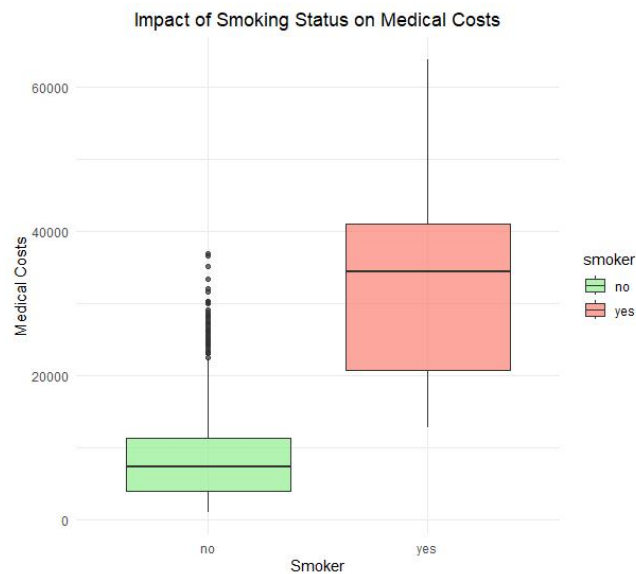
Table4. Correlation coefficient table

	age	bmi	children	charges
age	1	0.109272	0.042469	0.299008
bmi	0.109272	1	0.012759	0.198341
children	0.042469	0.012759	1	0.067998
charges	0.299008	0.198341	0.067998	1

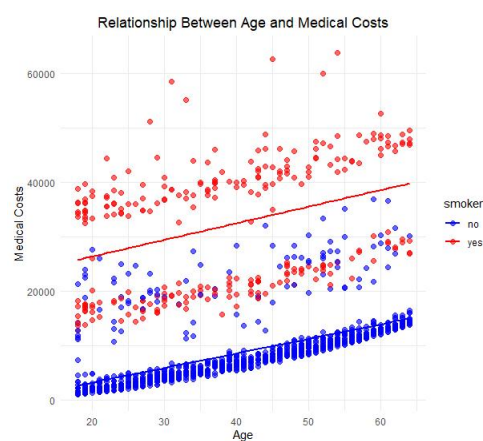
Calculate the Pearson correlation coefficients between age, BMI, number of children and medical expenses. The results show that the correlation between medical expenses and age is the strongest, while the correlations with BMI and the number of children are weaker. There is no strong multicollinearity among the variables.



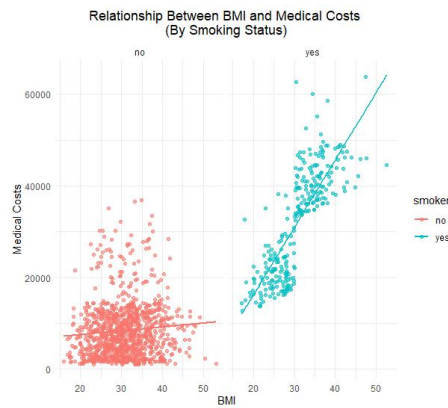
This histogram illustrates the distribution of medical expenses. The overall trend is a clear right skew, with most expenses concentrated in the lower range, while a few high expenses push up the tail of the distribution.



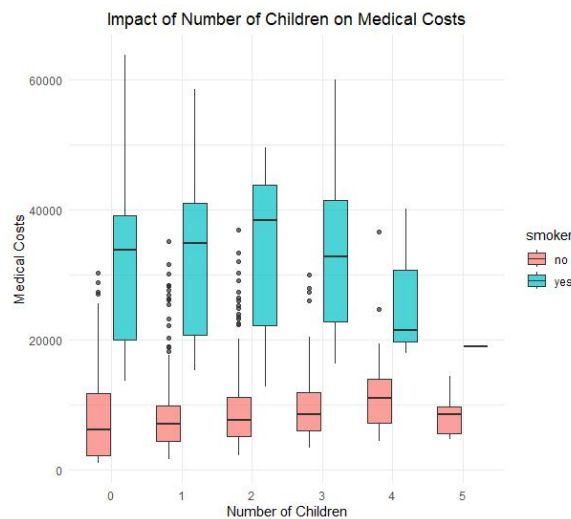
The box plot compares the distribution of medical expenses between smokers and non-smokers. The median and overall level of expenses for smokers are much higher than those of non-smokers. The position of the box has significantly shifted upwards, and the distribution range is also wider. The expenses of non-smokers are concentrated in the lower range, with only a few high-value outliers, which intuitively demonstrates the significant impact of smoking on medical expenditures.



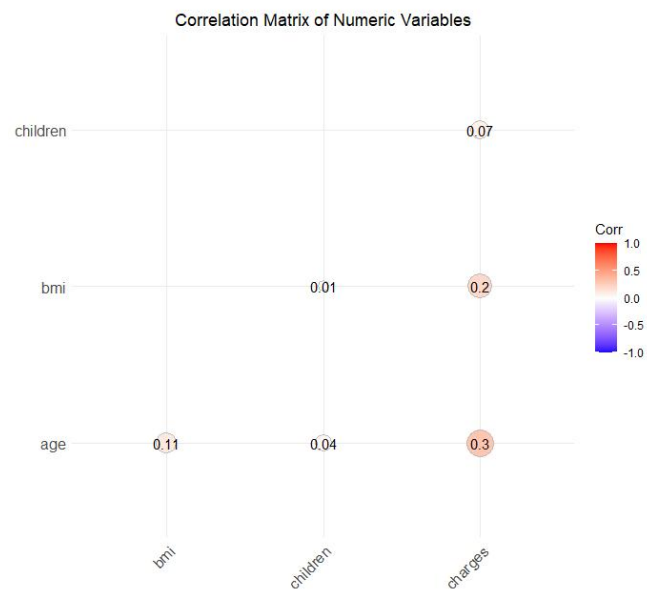
The scatter plot shows the relationship between age and medical expenses, with the data categorized by smoking status. Regardless of whether they smoke or not, medical expenses increase with age, and the overall expenses of smokers are significantly higher than those of non-smokers.



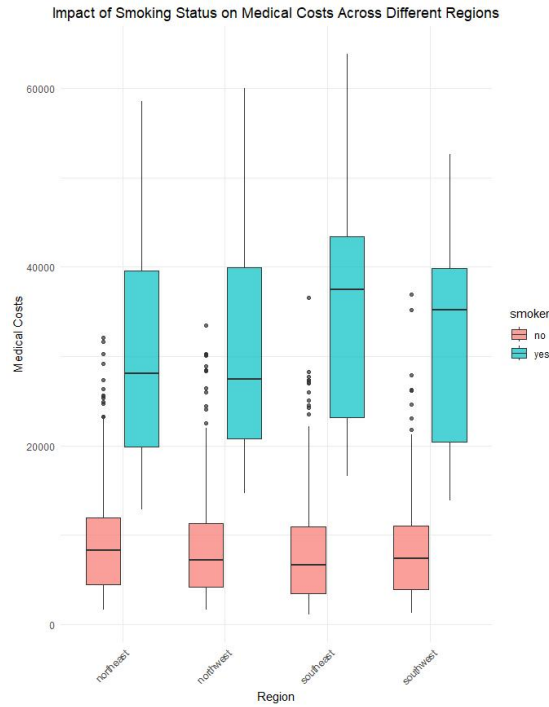
This graph is grouped by smoking status and shows the relationship between BMI and medical expenses. The expenses of smokers increase significantly as BMI rises, showing a strong positive correlation; the expenses of non-smokers are generally lower and have a weaker correlation with BMI.



The box plot compares the distribution of medical expenses for smokers and non-smokers across different numbers of children. The expenses of smokers are generally much higher than those of non-smokers, and the number of children has no significant impact on the expenses.



The correlation matrix heatmap illustrates the correlations among the numerical variables. Among them, the correlation between medical expenses and age is the strongest, while the correlations with BMI and the number of children are relatively weak.



The box plot compares the medical expenses of smokers and non-smokers in different regions. The medical expenses of smokers in each region are significantly higher than those of non-smokers. Among them, the median and distribution range of medical expenses for smokers in the southeastern region are the highest.

Methods

Before fitting the models, the categorical variables sex, smoker, and region were converted into factor variables. The dataset was then split into a training set and a testing set using a 70:30 ratio. The response variable was charges, and the predictor variables were age, sex, BMI, children, smoker, and region. All three models were trained on the training set, and predictions were made on the testing set. The random forest model was fitted with 500 trees. Model performance was compared using RMSE, MAE, and R-squared. Linear regression is a statistical technique used to

identify the relationship between variables. In the context of machine learning, linear regression is employed to discover the relationship between features and labels(Su, X.2012). In this study, the model was used to investigate the effects of factors such as age, gender, BMI, smoking status, number of children, and region on medical expenses. The coefficients were estimated using the least squares method, and the marginal effects of each variable on medical expenditures were quantified. The model assumes that the errors follow a normal distribution, that there is no multicollinearity among the independent variables, and that the error variance is homogeneous. The fitting effect can be evaluated by indicators such as R-squared. This model not only reveals the significant positive impact of key factors such as smoking and BMI on medical expenses, but also can be used to predict the medical expenditure levels of different characteristic populations, providing a reference basis for health management and insurance pricing.

Decision tree is a basic classification and regression method. The regression decision tree mainly refers to the CART (classification and regression tree) algorithm. The values of the internal node features are "yes" and "no", forming a binary tree structure. Regression means determining the corresponding output value based on the feature vector. The regression tree divides the feature space into several units, and each division unit has a specific output. Since each node is a judgment of "yes" or "no", the boundary of the division is parallel to the coordinate axis(Myles, A.2004). For test data, according to the features, it is classified into a certain unit, and the corresponding output value is obtained. The research uses decision trees to analyze

the influence of factors such as age, BMI, and smoking status on medical expenses. It automatically discovers the interaction relationships among variables. Compared with linear regression, it can better capture nonlinear correlations and is suitable for predicting complex real data and exploring factors.

Random forest is an ensemble learning algorithm based on decision trees. It constructs multiple decision trees by randomly selecting samples and features. For regression tasks, the final result is the average of all tree predictions, while for classification tasks, a voting mechanism is adopted. It can effectively reduce the risk of overfitting of a single decision tree, has strong noise resistance and good generalization ability, can handle both continuous and categorical variables, and can output variable importance. It is suitable for analyzing and modeling the influencing factors of medical expenses(Salman, H.2024).

The study selected the following three indicators to evaluate the performance of the model. RMSE (Root Mean Square Error) measures the standard deviation of the prediction error, with smaller values indicating higher accuracy; MAE (Mean Absolute Error) reflects the average level of the error, and the lower the value, the better; R-squared (Coefficient of Determination) represents the degree to which the model explains the variation in the data, and the closer it is to 1, the better the fitting effect.

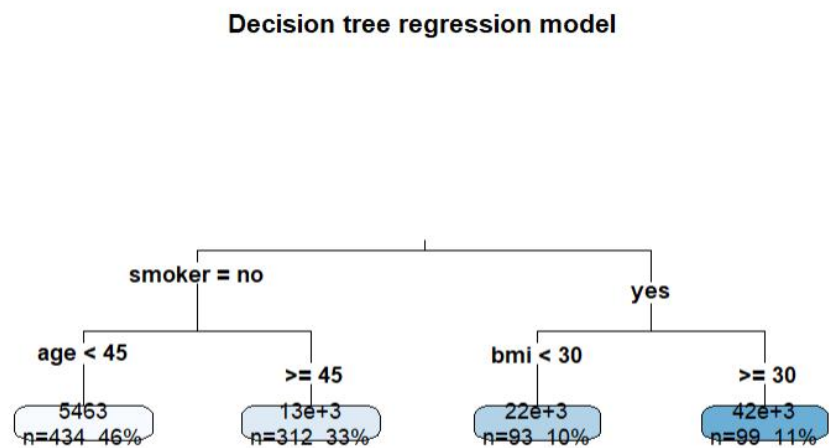
Results

Table5. Regression model result table

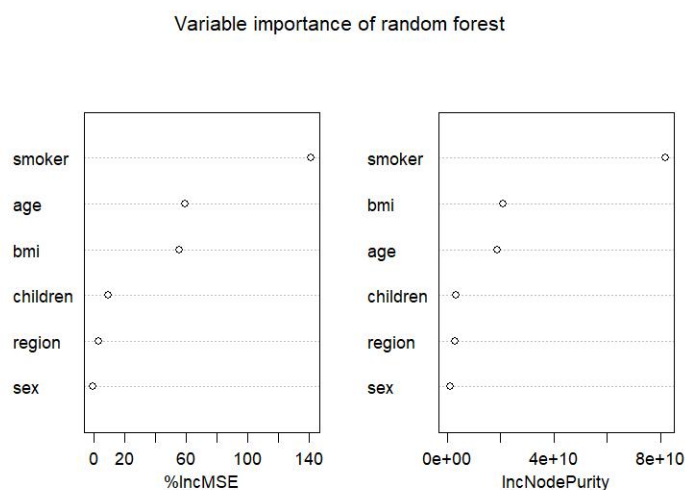
	Estimate	Standard Error	t value	Pr(> t)	
(Intercept)	-11,777.506	1,182.644	-9.959	0.0000	***
age	258.290	14.616	17.671	0.0000	***
sexmale	-401.261	407.214	-0.985	0.3247	
bmi	333.742	34.705	9.616	0.0000	***
children	397.245	170.165	2.334	0.0198	*
smokeryes	23,953.988	503.310	47.593	0.0000	***
regionnorthwest	-200.503	583.331	-0.344	0.7311	
regionsoutheast	-939.040	585.597	-1.604	0.1092	
regionsouthwest	-675.845	588.688	-1.148	0.2512	
Signif. codes: 0 <= '***' < 0.001 < '**' < 0.01 < '*' < 0.05					
Residual standard error: 6193 on 929 degrees of freedom					
Multiple R-squared: 0.746, Adjusted R-squared: 0.7438					
F-statistic: 341 on 929 and 8 DF, p-value: 0.0000					

The results of the linear regression model fitting show that smoking, age, BMI and the number of children have a significant positive impact on medical expenses, while gender and region have no significant effect. The adjusted R-square of the

model is 0.74, indicating a good overall fitting effect, and the p-value is much less than 0.001.



The decision tree regression model takes the smoking status as the primary splitting node. Among smokers, they are further divided by BMI, while non-smokers are classified by age. The results show that smoking is the core factor influencing medical expenses. The medical expenses of those with high BMI and who smoke are the highest, while the expenses of young non-smokers are the lowest.



The variable importance graph of the random forest shows through two indicators that the smoking status has the most significant impact on medical expenses,

with its importance being much higher than that of other variables. Next come age and BMI, while the effects of the number of children, region, and gender are relatively weak, which is highly consistent with the previous model analysis results.

Table6. Model Evaluation Comparison Table

Model	RMSE	MAE	R2
Linear Regression	5757.08	3967.68	0.7624
Decision Tree	4917.5	3212.57	0.8266
Random Forest	4400.76	2677.47	0.8612

The table compares the prediction performance of the three models on the test set, using three core indicators: RMSE, MAE and R-squared for evaluation.

The results show that the random forest model performs the best, with an RMSE of 4400.76, an MAE of 2677.47, and an R-squared of 0.8612. The decision tree model comes second, while the linear regression model performs the weakest. It can be seen that the random forest model effectively improves prediction accuracy and fitting goodness by reducing the risk of overfitting. This indicates that there is a strong nonlinear relationship between medical expenses and related factors. The random forest model can better capture complex interaction effects and is more suitable for this prediction scenario.

Discussion of Results

The three models show different strengths and weaknesses. Linear regression is easy to interpret and clearly shows the effect of each predictor, but it may not fully capture nonlinear relationships among smoking status, BMI, age, and medical expenses. The decision tree model is easier to visualize and can capture some nonlinear patterns, but it may be less stable and more likely to overfit than ensemble methods. The random forest model has the best predictive performance because it combines many decision trees and reduces the risk of overfitting. However, it is less interpretable than linear regression.

Based on the test set results, the random forest model is recommended as the final model because it has the lowest RMSE and MAE and the highest R-squared. From a practical perspective, smoking status, age, and BMI should be considered the most important factors when predicting individual medical expenses.

Conclusion

Through descriptive analysis and three predictive models, this study explored the key influencing factors of individual medical expenses. The results indicated that smoking status, age, and BMI were the core variables affecting medical expenses. Among them, the medical expenditures of smokers were significantly higher, and the increase in BMI and age would further raise the costs. The influences of gender, region, and the number of children were relatively limited. Model comparisons

showed that random forest outperformed decision trees and linear regression in terms of prediction accuracy and error control, and was better at capturing the nonlinear relationships between variables. In conclusion, health interventions targeting high smoking rates and high BMI populations can effectively reduce medical expenses, providing data support for health management and insurance pricing.

References

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